Exam Date & Time: 20-Nov-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

III Semester B.Tech Biomedical Engineering
End Semester Examinations - November 2024

ELECTRONICS CIRCUITS [BME 2122]

Marks: 50 **Duration: 180 mins.**

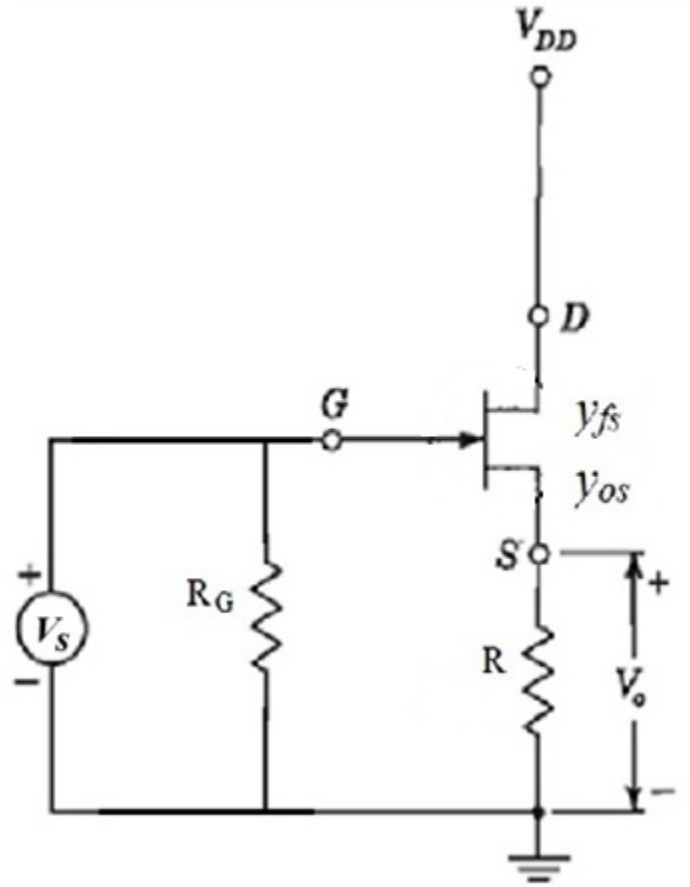
Descriptive

Answer all the questions. Section Duration: 180 mins

Draw diagrams wherever necessary
Assume suitable missing data if any

1A) Construct a sinusoidal oscillator using a JFET amplifier and frequency selective network and explain its working in detail. (4)

1B) Analyze the following amplifier for input impedance, output impedance and voltage gain and draw inferences from the analysis. (4)



1C) How do you make use of op-amp zero-cross detector as a sinusoidal to square wave converter? Explain. (2)

2A)	Build an astable multivibrator using an operational amplifier and explain its working in detail.	(5)
2B)	Design an op-amp inverting Schmitt trigger such that V_h = 7V and maximum output swing is ±14 V.	(3)
2C)	With relevant circuit diagrams, give justification for the existence of virtual ground in an inverting op-amp configuration.	(2)
3A)	How do you generate free-running ramp signal using 555 timer? Explain.	(5)
3B)	Making use of 555 timer IC and appropriate external timing components design an astable multivibrator to generate a square of 2KHz frequency and 50% duty cycle.	(3)
3C)	Write any four applications of 555 timer.	(2)
4A)	Construct a voltage-controlled oscillator using 555-timer astable multivibrator and write its applications.	(3)
4B)	How do you obtain line regulation characteristics of IC 7805 voltage regulator? Explain.	(3)
4C)	Design a voltage regulator using IC 7805 for the following Specifications: V_O =10V, I_L = 400mA. Assume a quiescent current of 4.3 mA for the voltage regulator IC.	(4)
5A)	Construct a 4-bit R-2R ladder DAC using an Op-amp and explain its operation.	(4)
5B)	Making use of appropriate components, construct a 4-bit parallel comparator ADC and explain its operation. Assume V_{REF} = 5V.	(4)
5C)	If a 4-bit DAC generates 0.2V for the digital input 0010, compute the output for digital inputs: (i) 1010 (ii) 1100	(2)

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